

Ashutosh Kumar

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🌐 github.com/Ashu411 | 🔗 [LinkedIn](#) | 📄 [LeetCode](#) | 🌐 [Portfolio](#)



PROFESSIONAL SUMMARY

Aspiring Software Engineer with expertise in Python, Data Structures Algorithms, and Machine Learning. Experienced in building AI/ML applications, predictive analytics solutions, and data-driven software using modern development practices. Passionate about solving complex problems, continuous learning, and delivering impactful technology solutions.

EDUCATION

Jaypee Institute of Information Technology (JIIT)
Bachelor of Technology in Computer Science and Engineering

Aug 2024 – May 2028
Noida, Uttar Pradesh

- CGPA: 7.0 / 10
- Relevant Coursework: DSA, OOP, DBMS, Machine Learning

Delhi Public School
Class XII — CBSE

Apr 2022 – Mar 2023
New Delhi

- Aggregate: **90%**

EXPERIENCE

Codec Technologies
Artificial Intelligence Intern

June 2026 – August 2026
Remote

- Developed end-to-end Machine Learning pipelines using Python, Pandas, NumPy, and Scikit-learn for data preprocessing, feature engineering, model development, and evaluation.
- Performed exploratory data analysis (EDA), data cleaning, feature encoding, and scaling to prepare datasets for supervised learning models.
- Implemented and evaluated classification models using precision, recall, and F1-score to support model selection and performance analysis.
- Collaborated using Git and GitHub for version control, code reviews, and documentation while following software engineering best practices.

Verification: [Offer Letter](#)

TECHNICAL SKILLS

Languages	C++, Python, Java, C, JavaScript
AI / ML	TensorFlow, Scikit-Learn, OpenCV, CNNs, Deep Learning, Pandas, NumPy
Web Dev	React.js, Node.js, HTML5, CSS3, REST APIs
Databases	MySQL, Git, GitHub
Tools & Platforms	Linux, VS Code, Overleaf, Jupyter Notebook
Soft Skills	Strategic Leadership, Analytical Thinking, Cross-functional Collaboration
Languages	English (Fluent), Hindi (Fluent)

PROJECTS

Customer Churn Prediction using Machine Learning | *Python · Pandas · NumPy · Scikit-learn · XGBoost · Random Forest · [GitHub](#) Matplotlib · Seaborn*

- Built an end-to-end ML pipeline for customer churn prediction covering data cleaning, feature engineering, model training, and evaluation using XGBoost, Random Forest, and Logistic Regression.
- Conducted thorough EDA to surface churn-driving patterns; applied label encoding, feature scaling, and class-imbalance handling to improve model robustness across classification metrics.
- Compared model performance via precision, recall, and F1-score; visualised feature importances and churn distributions using Seaborn and Matplotlib to generate actionable retention insights.

Twitter Sentiment Analysis using NLP | *Python · NLTK · Pandas · Scikit-learn · TF-IDF · Logistic Regression*

[GitHub](#)

- Developed an NLP-based sentiment classification pipeline for Twitter data; implemented tokenization, stopword removal, and lemmatization using NLTK for robust text preprocessing.
- Applied TF-IDF vectorization to transform cleaned tweets into numerical features, then trained and evaluated a Logistic Regression classifier; analysed results via confusion matrix and classification report.

Tomato Leaf Disease Detector | *Python · TensorFlow · CNN · OpenCV · NumPy*

[GitHub](#)

- Engineered an end-to-end AI-based image classification pipeline to detect diseases in tomato leaves, achieving high-accuracy multi-class predictions using a custom Convolutional Neural Network (CNN).
- Optimised training data quality by implementing standardised preprocessing workflows — including image resizing, normalisation, and augmentation — reducing model overfitting and improving validation accuracy.

- Designed and trained a multi-layer CNN architecture from scratch, selecting hyperparameters and activation functions to balance precision and recall for both healthy and diseased leaf categories.
- Integrated OpenCV for real-time image input handling, enabling the model to process raw field images and return actionable disease classifications.

ShramSetu — Labour Management Web Platform | *React.js · Node.js · JavaScript · HTML · CSS* [GitHub](#)

- Architected a full-stack, role-based labour management platform supporting secure multi-user authentication with distinct workflows for labourers, supervisors, and administrators.
- Implemented OTP-based attendance verification and real-time check-in/check-out tracking, reducing manual attendance errors and enabling accurate automated wage calculation.
- Developed a responsive dashboard providing supervisors with live visibility into work hours, payment verification status, and labour welfare metrics, streamlining operational oversight.
- Designed RESTful backend workflows in Node.js for role-gated access control, ensuring data integrity and secure payroll processing across user tiers.

Sales Performance Gap Analysis — Benchmark-Based Analytics | *Python · Pandas · Seaborn · EDA · Data Visualisation* [GitHub](#)

- Built a benchmark-driven analytics model to systematically identify underperforming products by comparing actual sales against defined KPI thresholds across product categories.
- Conducted comprehensive data cleaning and exploratory data analysis (EDA) on multi-dimensional sales datasets, uncovering hidden performance gaps that informed strategic decision-making.
- Generated actionable visual reports using Seaborn, translating complex analytical findings into clear business insights that reduced manual reporting effort and improved decision cycle time.

Pandemic Spread Simulation & Containment Analysis | *C++ · Graph Theory · Probabilistic Modelling* [GitHub](#)

- Developed a graph-theory-based pandemic spread simulation in C++, modelling inter-city disease transmission using probabilistic infection models and adjacency-list graph representations.
- Simulated multiple containment strategies — including partial lockdowns and quarantine zones — enabling comparative analysis of infection curve suppression under different policy scenarios.
- Delivered real-time statistics output (infection rates, recovery curves, transmission hotspots), demonstrating the practical application of algorithmic problem-solving to epidemiological modelling.

CERTIFICATIONS

Generative AI Certification — TechBlocks 101, IEEE SB JIIT	2024
Generative AI fundamentals, LLM concepts, prompt engineering, AI application design.	
Data Analytics Job Simulation — Deloitte (Forage)	2026
End-to-end analytics workflow: data wrangling, interpretation, and business insight generation.	
AI Tools & ChatGPT Workshop — be10X	2026
Practical generative AI, prompt engineering strategies, and AI-driven productivity workflows.	
Technology Job Simulation — Deloitte (Forage)	2026
Software development lifecycle, structured problem-solving, and technology consulting exposure.	

ACHIEVEMENTS & ACTIVITIES

- Active contributor to open-source repositories on GitHub; maintained projects with documented READMEs, clean commit histories, and modular codebases.
- Consistently practising Data Structures & Algorithms on LeetCode, with focus on dynamic programming, graphs, and tree-based problems.
- Participated in IEEE SB JIIT TechBlocks 101 programme, engaging with industry experts on Generative AI and Large Language Model applications.
- Completed two Deloitte Forage job simulations (Data Analytics + Technology) in 2026, gaining exposure to professional consulting and analytics workflows.